Statistical Computing HW1

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Contents

1 Weibull distribution: 2

1.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf. 2

1.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue. 3

2 Pareto distribution: 3

2.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf. 3

2.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue. 5

3 Skewed Distribution I: 5

3.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf. 5

3.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue. 9

4 Skewed Distribution II: 10

4.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf. 10

4.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue. 13

5 A 2-dimension distribution: 14

5.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf. 14

5.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue. 18

6 Acknowledgement 19

7 Appendix **19**

7.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf. 19

7.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue. 25

1 Weibull distribution:

$$F(x) = 1 - e^{-(x/\theta)^\beta}, \quad \theta > 0,\ \beta > 0,\ x > 0.$$

1.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf.

By the following derivation, it is fairly easy to show that $F^{-1}(x) = \theta[-\log(1-x)]^{1/\beta}$ for $0 < x < 1$.

$$\begin{aligned}
y &\overset{Let}{=} F(x) = 1 - e^{-(x/\theta)^\beta} \\
&\implies 1 - y = e^{-(x/\theta)^\beta} \\
&\implies -\log(1-y) = (x/\theta)^\beta \\
&\implies \theta[-\log(1-y)]^{1/\beta} = x \\
&\implies F^{-1}(x) = \theta[-\log(1-x)]^{1/\beta}, \quad 0 < x < 1
\end{aligned}$$

Set the parameter values $\theta = 1$ and $\beta = 3$. We thus have $[-\log(1-x)]^{1/3}$.

1. Algorithm in detail steps

Inverse cdf method is applied. The steps are detailed as follows.

step1: Generate 1000 random numbers from $U(0,1)$.

step2: Transform the random numbers in 2. via $F^{-1}(\cdot)$

2. Draw a sample of size 1000.

```
F.inv = function(x){(-log(1-x))^(1/3)}
```

```
x.vec = F.inv(runif(1000))
```

3. compare the empirical distribution of your data to the target pdf.

Figure 1 shows that the empirical pdf looks very close to the target pdf.

```
hist(x.vec, freq = FALSE, ylim = c(0, 1.2), main = "", xlab = expression(x))
x = seq(0,10, length.out = 1000)
lines(x, dweibull(x, shape = 3, scale = 1), col = "red")
legend("topright", "target pdf", lty = 1, col = "red")
```

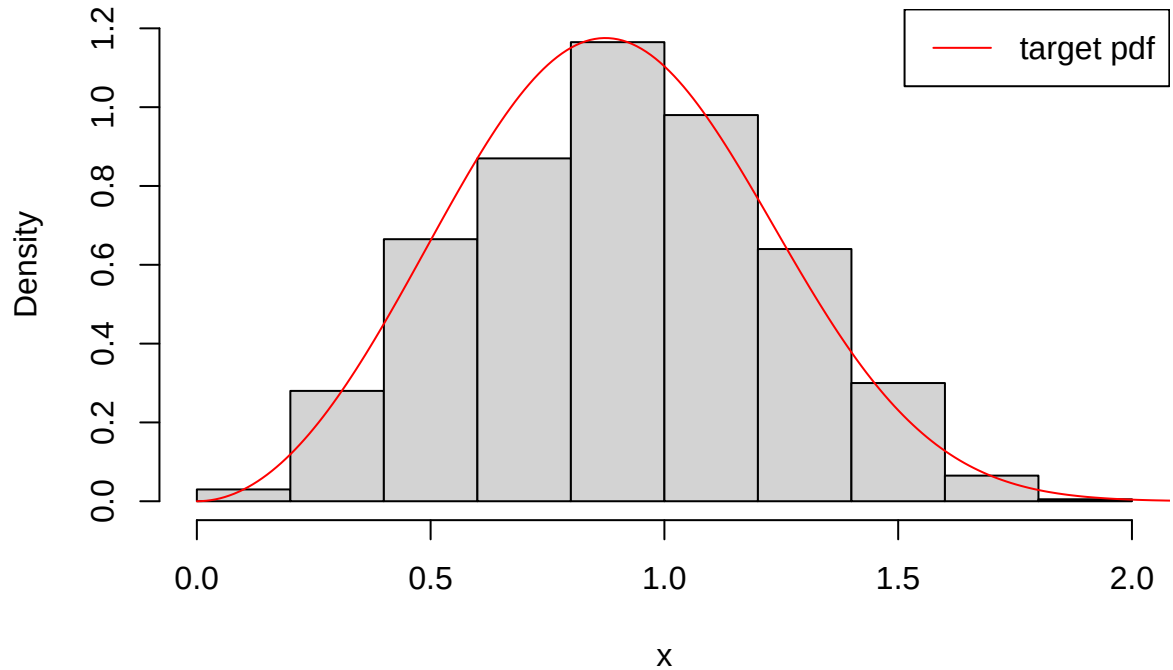


Figure 1: Empirical vs target

1.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue.

The efficiency is 100% both theoretically and empirically since the inverse cdf method used would not involve any waste of sample in essence. Also the efficiency would not change with the parameter values since it has nothing to do with the setting of parameter values.

2 Pareto distribution:

$$f(x) = \frac{\beta}{\theta(1 + x/\theta)^{\beta+1}}, \quad \theta > 0,\ \beta > 0,\ x > 0.$$

2.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf.

By the following derivation, it is fairly easy to show that $F^{-1}(x) = 1 - \frac{1}{(1+x/\theta)^{\beta}}$ for $0 < x < 1$.

$$
\begin{aligned}
F(x) &= \int_0^x \frac{\beta}{\theta(1+y/\theta)^{\beta+1}} dy \\
&\overset{Let\, t=y/\theta}{=} \int_0^{x/\theta} \frac{\beta}{(1+t)^{\beta+1}} dt \\
&= -\frac{1}{(1+t)^{\beta}}\Big|_0^{x/\theta} \\
&= 1 - \frac{1}{(1+x/\theta)^{\beta}} \overset{Let}{=} y \\
&\implies 1 + x/\theta = \left(\frac{1}{1-y}\right)^{1/\beta} \\
&\implies x = \theta\left[\left(\frac{1}{1-y}\right)^{1/\beta} - 1\right] \\
&\implies F^{-1}(x) = \theta\left[\left(\frac{1}{1-y}\right)^{1/\beta} - 1\right], \quad 0 < x < 1
\end{aligned}
$$

Set the parameter values $\theta = 1$ and $\beta = 3$. We thus have $\left(\frac{1}{1-x}\right)^{1/3} - 1$.

1. Algorithm in detail steps

Inverse cdf method is applied. The steps are detailed as follows.

step1: Generate 1000 random numbers from $U(0, 1)$.

step2: Transform the random numbers in 2. via $F^{-1}(\cdot)$

2. Draw a sample of size 1000**.**

```
F.inv = function(x){(1/(1-x))^{1/3}-1}
```

```
x.vec = F.inv(runif(1000))
```

3. compare the empirical distribution of your data to the target pdf.

Figure 2 shows that the empirical pdf looks very close to the target pdf.

```
dpar = function(x, theta = 1, beta = 3){
  beta/(theta*(1+x/theta)^(beta+1))
}
hist(x.vec, freq = FALSE, main = "", xlab = expression(x))
x = seq(0,10, length.out = 1000)
lines(x, dpar(x), col = "red")
legend("topright", "target pdf", lty = 1, col = "red")
```

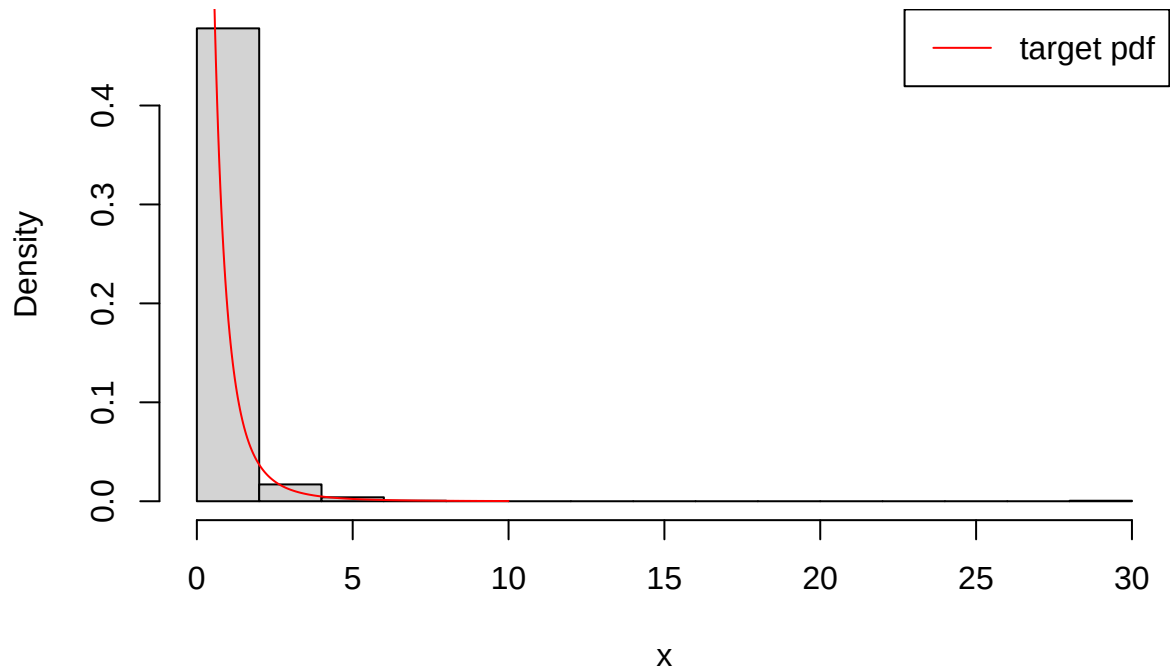


Figure 2: Empirical vs target

2.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue.

The efficiency is 100% both theoretically and empirically since the inverse cdf method used would not involve any waste of sample in essence. Also the efficiency would not change with the parameter values since it has nothing to do with the setting of parameter values.

3 Skewed Distribution I:

$$f(x) = \frac{2}{\gamma + \frac{1}{\gamma}} \phi(x\gamma^{-sign(x)}), \quad \gamma > 0,\, x > 0,$$

where $\phi(x)$ is the pdf of $N(0,1)$. The parameter γ controls the degree of asymmetry. In particular, $f(x)$ becomes symmetric when $\gamma = 1$.

3.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf.

1. Find a suitable proposal distribution

Noting that

$$f(x) = \begin{cases} \frac{2}{\gamma+\frac{1}{\gamma}}\phi(\frac{x}{\gamma}), & x < 0, \\ \frac{2}{\gamma+\frac{1}{\gamma}}\phi(x), & x = 0, \\ \frac{2}{\gamma+\frac{1}{\gamma}}\phi(x\gamma), & x > 0, \end{cases}$$

we consider a proposal distribution with density function $g(x) = \frac{1}{\max(\gamma,\frac{1}{\gamma})}\phi\Big(\frac{x}{\max(\gamma,\frac{1}{\gamma})}\Big)$ so that it could have tails heavier than those of the target one on both sides up to multiplication of a constant. By solving the following optimization problem, we can derive $M(\gamma)$ such that $f(x) \le M(\gamma)g(x)$ for all $x \in R$. Such $M(\gamma)$ can be shown optimal for each $\gamma > 0$ in the sense that it provides the highest efficiency for the algorithm.

$$\begin{aligned} M(\gamma) = \max_{x\in R}\frac{f(x)}{g(x)} &= \max_{x\in R}\frac{\frac{2}{\gamma+\frac{1}{\gamma}}\phi(x\gamma^{-sign(x)})}{\frac{1}{\max(\gamma,\frac{1}{\gamma})}\phi\Big(\frac{x}{\max(\gamma,\frac{1}{\gamma})}\Big)} \\ &= \max_{x\in R}\begin{cases} \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}e^{-\frac{1}{2}(\gamma^2-\max(\gamma,\frac{1}{\gamma})^{-2})x^2}, & x \le 0 \\ \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}e^{-\frac{1}{2}(\gamma^{-2}-\max(\gamma,\frac{1}{\gamma})^{-2})x^2}, & x > 0. \end{cases} \end{aligned}$$

Let's then discuss on two cases: $0 < \gamma \le 1$ and $\gamma > 1$.

1. *Case 1:* $0 < \gamma \le 1$*:*

$$\begin{aligned} M(\gamma) &= \max_{x\in R}\begin{cases} \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}e^{-\frac{1}{2}(\gamma^2-\gamma^2)x^2}, & x \le 0 \\ \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}e^{-\frac{1}{2}(\gamma^{-2}-\gamma^2)x^2}, & x > 0. \end{cases} \\ &= \max_{x\in R}\begin{cases} \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}, & x \le 0 \\ \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}e^{-\frac{1}{2}(\gamma^{-2}-\gamma^2)x^2}, & x > 0. \end{cases} \\ &= \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}, \quad 0 < \gamma \le 1. \end{aligned}$$

The last equality follows since $e^{-\frac{1}{2}(\gamma^{-2}-\gamma^2)x^2} \le 1$ when $0 < \gamma \le 1$

2. *Case 2:* $\gamma > 1$

$$\begin{aligned} M(\gamma) &= \max_{x\in R}\begin{cases} \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}e^{-\frac{1}{2}(\gamma^2-\gamma^{-2})x^2}, & x \le 0 \\ \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}e^{-\frac{1}{2}(\gamma^{-2}-\gamma^{-2})x^2}, & x > 0. \end{cases} \\ &= \max_{x\in R}\begin{cases} \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}e^{-\frac{1}{2}(\gamma^2-\gamma^{-2})x^2}, & x \le 0 \\ \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}, & x > 0. \end{cases} \\ &= \frac{2\max(\gamma,\frac{1}{\gamma})}{\gamma+\frac{1}{\gamma}}, \quad \gamma > 1. \end{aligned}$$

The last equality follows since $e^{-\frac{1}{2}(\gamma^2-\gamma^{-2})x^2} \le 1$ when $\gamma > 1$.

Set the parameter values $\gamma = 2$. Then the results above suggest that $M(2) = 1.6$ and $g(x) = \frac{1}{2}\phi(\frac{x}{2})$. Figure 3 shows that $f(x) \le Mg(x)$ for all $x \in R$ when $M = 1.6$.

```r
f = function(x, gamma = 2){
  (2/(gamma+1/gamma))*dnorm(x*gamma^(-sign(x)))
}
g = function(x, gamma = 2){
  dnorm(x, 0, max(gamma, 1/gamma))
}
```

```r
curve(f(x), type = "l", xlab = expression(x), ylab = expression(f(x)), from = -10, to = 10)
curve(g(x), lty = 2, col = 2, add = TRUE)
M = (2*max(2,1/2))/(2+1/2); curve(g(x)*M, lty = 1, col = 4, add = TRUE);
legend("topright", c(expression(f(x)), expression(g(x)), expression(M%*%g(x))),
                      col = c(1, 2, 4), lty = c(1, 2, 1))
```

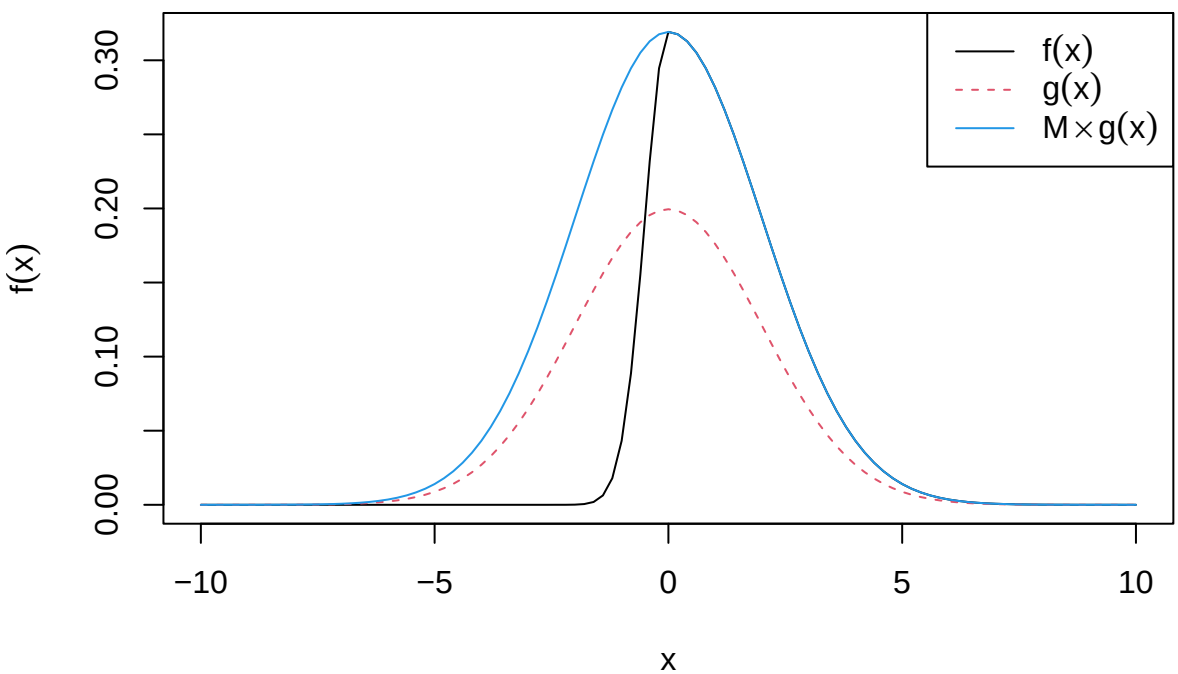


Figure 3: Plot for the densities

2. Algorithm in details

Rejection sampling is applied. The steps are detailed as follows.

step1: Generate $Y \sim g$ and $U \sim U(0,1)$.

step2: Let $X = Y$ if $U \leq \frac{f(Y)}{Mg(Y)}$, and reject the draw otherwise.

step3: Iterate step1 and step2 until 1000 $X's$ are generated throughout.

3. Draw a sample of size 1000**.**

```
x.vec = c()
n.gen = 0
gamma = 2
set.seed(111024513)
while(length(x.vec)<1000){
  x = rnorm(1, 0, max(gamma, 1/gamma))
  if(runif(1) <= f(x)/(M*g(x))){
    x.vec = c(x.vec, x)
  }
   n.gen = n.gen + 1
}
```

4. compare the empirical distribution of your data to the target pdf.

Figure 4 shows that the empirical pdf looks very close to the target pdf.

```
hist(x.vec, xlim  = c(-10, 10), main = "", freq = FALSE, xlab = expression(x), ylim = c(0, 0.32))
curve(f(x), col = "red", add = TRUE)
legend("topright", "target pdf", lty = 1, col = "red")
```

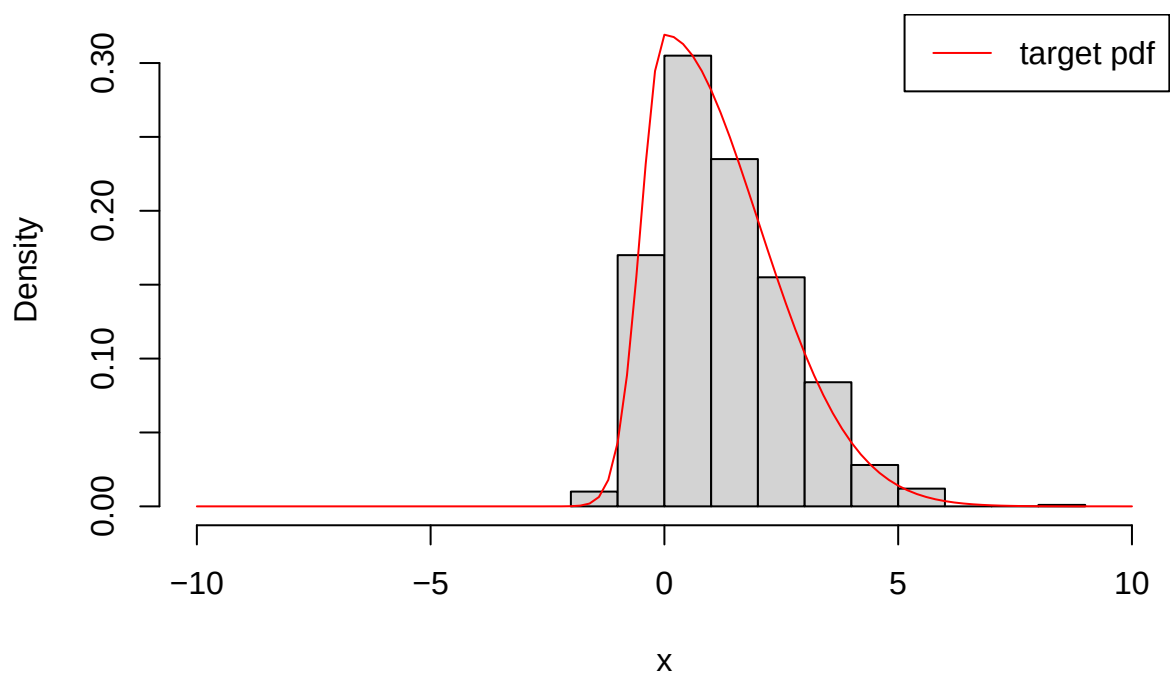


Figure 4: Empirical vs target

3.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue.

```
1000/n.gen
```

```
## [1] 0.6218905
```

1. The theoretical efficiency is $\frac{1}{M} = 62.5$ % and the empirical efficiency can be obtained by dividing 1000 to the number of the random numbers generated from $g(\cdot)$. It turns out that the empirical efficiency is 62.1891 %. We see that the empirical efficiency is very close to the theoretical one as expected.

2. Since the formula for calculating theoretical efficiency is $1/M(\gamma)$ is a function of *gamma*. The efficiency will change with the parameter value γ. Figure 5 shows how the efficiency would change when varying the value of γ.

```
eff = function(x){
  v = rep(NA, length(x))
  for(i in 1:length(v)){
    v[i] = 1/((2*max(x[i], 1/x[i]))/(x[i]+1/x[i]))
  }
  return(v)
}
gamma = seq(10^(-5), 3, length.out = 1000)
plot(gamma, eff(gamma), type = "l", xlab = expression(gamma), ylab = "efficiency")
abline(v = 1, col = "red")
```

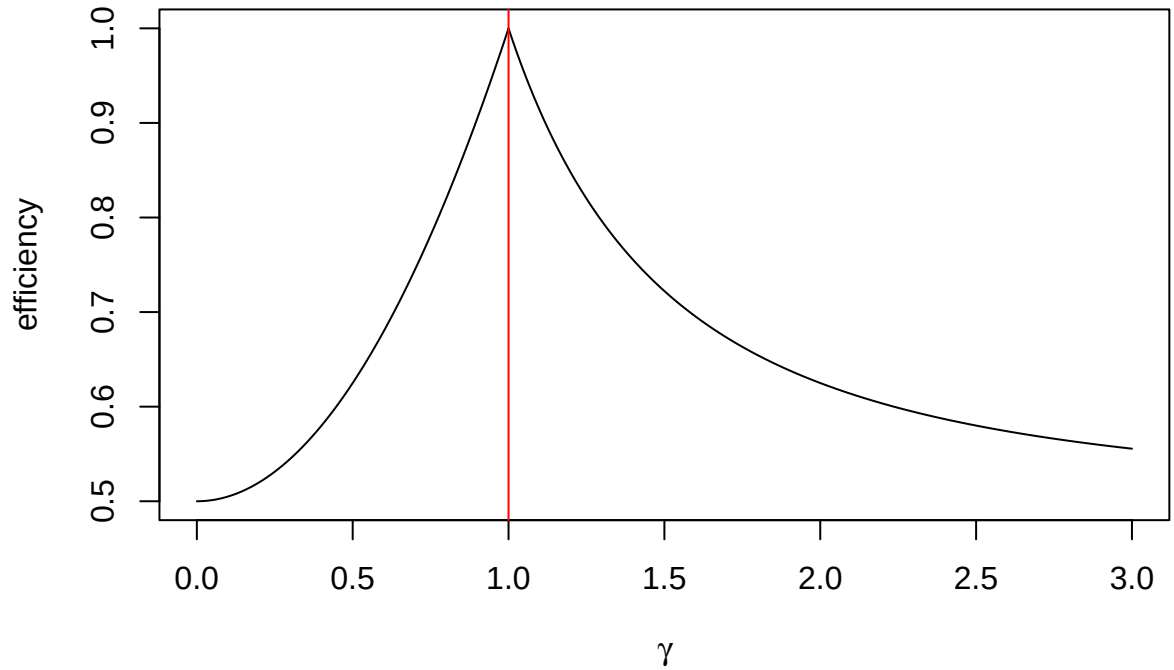


Figure 5: One can see that the efficiency increases as a function of γ when γ <1 and decreases along γ when γ exceeds one.

4 Skewed Distribution II:

$$f(x) = 2h(x)G(\alpha x), \quad x \in R,\ \alpha \in R,$$

where $\phi(x)$ is the pdf of $N(0,1)$. The parameter γ controls the degree of asymmetry. In particular, $f(x)$ becomes symmetric when $\gamma = 1$.

If $G'(x) = h(x)$, $f(x)$ is called the skew-"G" distribution. For example, when $G(\cdot)$ and $h(\cdot)$ are cdf and pdf of the same normal distribution, $f(\cdot)$ is called skew-normal distribution. When $G(\cdot)$ and $h(\cdot)$ are cdf and pdf of the t-distribution, $f(x)$ is called skew-t distribution. In this homework, you need to generate random samples from this skew-t distribution with parameter (α, ν).

4.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf.

1. Find a suitable proposal distribution

To make things more clear, rewrite $f(x)$ as

$$f(x) = 2h_\nu(x)G_\nu(\alpha x), \quad x \in R,\ \alpha \in R,$$

so that it might become more obvious to recognize $f(x)$ as a function of α and ν.

Consider a proposal distribution with density function $g(x) = h_\nu(x)$ so that it could have tails heavier than those of the target one on both sides up to multiplication of a constant. By solving the following optimization problem, we can derive $M(\nu, \alpha)$ such that $f(x) \leq M(\nu, \alpha)g(x)$ for all $x \in R$. Such $M(\nu, \alpha)$ can be shown optimal for each $\gamma > 0$ in the sense that it provides the highest efficiency for the algorithm.

$$M(\alpha, \nu) = \max_{x \in R} \frac{f(x)}{g(x)} = \max_{x \in R} \frac{2h_\nu(x)G_\nu(\alpha x)}{h_\nu(x)} = \max_{x \in R} 2G_\nu(\alpha x) = 2$$

The last equality follows easily by noting that $G_\nu(\cdot)$ is a cdf function.

Set the parameter values $\alpha = 3$ and $\nu = 8$. Then the results above suggest $M(3, 8) = 2$ and $g(x)$ to be the pdf of t-distribution with degree of freedom equal to 8. Figure 6 shows that $f(x) \leq Mg(x)$ for all $x \in R$ when $M = 2$.

```
f = function(x, alpha = 3, nu = 8){
  2*dt(x, df = nu)*pt(x, df = alpha*nu)
}
g = function(x, nu = 8){
  dt(x, df = nu)
}
```

```
curve(f(x), type = "l", xlab = expression(x), ylab = expression(f(x)),
      from = -10, to = 10, ylim = c(0, 0.8))
curve(g(x), lty = 2, col = 2, add = TRUE)
M = 2; curve(g(x)*M, lty = 1, col = 4, add = TRUE);
legend("topright", c(expression(f(x)), expression(g(x)), expression(M%*%g(x))),
                     col = c(1, 2, 4), lty = c(1, 2, 1))
```

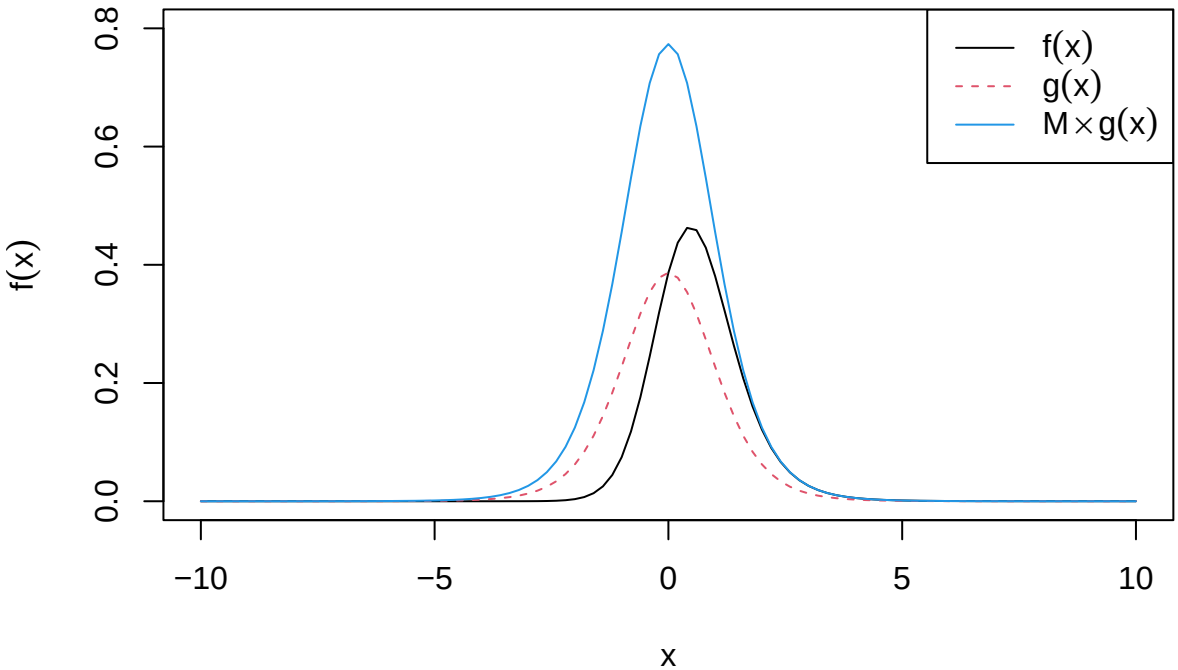


Figure 6: Plot for the densities

2. Algorithm in details

Rejection sampling is applied. The steps are detailed as follows.

step1: Generate $Y \sim g$ and $U \sim U(0,1)$.

step2: Let $X = Y$ if $U \leq \frac{f(Y)}{Mg(Y)}$, and reject the draw otherwise.

step3: Iterate step1 and step2 until 1000 $X's$ are generated throughout.

3. Draw a sample of size 1000**.**

```
x.vec = c()
n.gen = 0
nu = 8
set.seed(111024513)
while(length(x.vec)<1000){
  x = rt(1, df = nu)
  if(runif(1) <= f(x)/(M*g(x))){
    x.vec = c(x.vec, x)
  }
   n.gen = n.gen + 1
}
```

4. compare the empirical distribution of your data to the target pdf.

Figure 7 shows that the empirical pdf looks very close to the target pdf.

```
hist(x.vec, xlim  = c(-10, 10), , ylim = c(0, 0.6), main = "", freq = FALSE, xlab = expression(x))
curve(f(x), col = "red", add = TRUE)
legend("topright", "target pdf", lty = 1, col = "red")
```

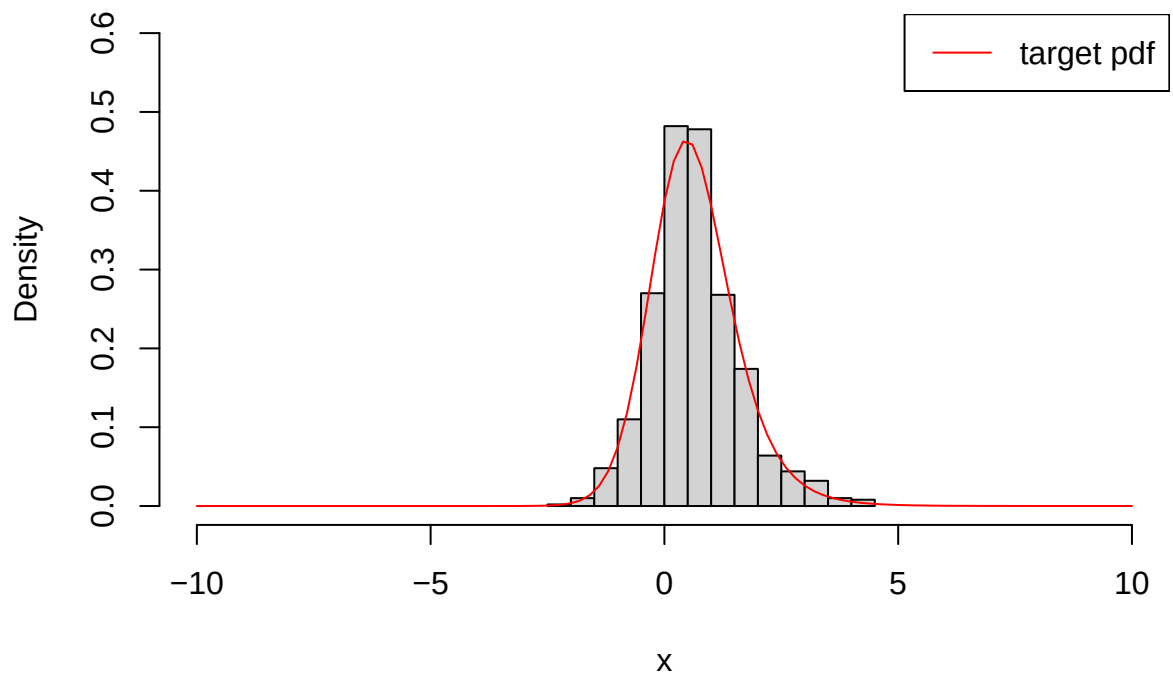


Figure 7: Empirical vs target

4.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue.

```
1000/n.gen
```

```
## [1] 0.5065856
```

1. The theoretical efficiency is $\frac{1}{M} = 50$ % and the empirical efficiency can be obtained by dividing 1000 to the number of the random numbers generated from $g(\cdot)$. It turns out that the empirical efficiency is 50.6586 %. We see that the empirical efficiency is very close to the theoretical one as expected.

2. The formula for calculating theoretical efficiency is $1/M(\nu, \alpha)$, which is a constant function taking value on 0.5. Therefore, the efficiency will never change and stays 0.5 for all $(\alpha, \nu) \in R \times R^+$.

5 A 2-dimension distribution:

$$f(x,y) = 2(1-x)(1-y)(1-xy)^{-3}, \quad 0 < x < 1,\, 0 < y < 1.$$

5.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf.

1. Find a suitable proposal distribution

Let $f(y) = 1$ and $f(x|y) = 2(1-x)(1-y)(1-xy)^{-3}$. It turns out that it is sufficient to find out how to generate random number from $f(x|y)$ since $f(y)$ is merely a density of uniform distribution. Consider $g(x) = 1, 0 < x < 1$, as a proposal distribution. We can find out an optimal $M(y)$ such that $f(x|y) \leq M(y)g(x)$ for all $0 < x < 1$ by first taking first derivative of $f(x|y)/g(x)$ with respect to x. The derivation is provided as follows.

$$\begin{aligned}
\frac{d}{dx}[f(x|y)/g(x)] &= \frac{d}{dx}2(1-x)(1-y)(1-xy)^{-3} \\
&= 2(1-y)\frac{-(1-xy)^3 + 3y(1-x)(1-xy)^2}{(1-xy)^6} \\
&= -2(1-y)\frac{(1-xy)^2[1-3y+2xy]}{(1-xy)^6} \overset{let}{=} 0 \\
&\implies 1-3y+2xy = 0 \\
&\implies x = \frac{3y-1}{2y}.
\end{aligned}$$

Furthermore, since

$$\frac{d}{dx}[f(x|y)/g(x)] = -2(1-y)\frac{(1-xy)^2[1-3y+2xy]}{(1-xy)^6}\begin{cases} > 0, & x < \frac{3y-1}{2y} \\ = 0, & x = \frac{3y-1}{2y} \\ < 0, & x > \frac{3y-1}{2y} \end{cases},$$

it follows that $f(x|y)/g(x)$ has maximum at $x = \frac{3y-1}{2y}$ for some given y. Note that $0 < x = \frac{3y-1}{2y} < 1 \implies \frac{1}{3} < y < 1$, which means the point of maximum just found is valid only for $\frac{1}{3} < y < 1$. For $\frac{1}{3} < y < 1$, since $\frac{d}{dx}[f(x|y)/g(x)] = -2(1-y)\frac{(1-xy)^2[1-3y+2xy]}{(1-xy)^6} < 0$, it follows that $f(x|y)/g(x)$ has maximum at $x = 0$ for $\frac{1}{3} < y < 1$. Therefore, we have

$$M(y) = \max_{0<x<1}\frac{f(x|y)}{g(x)} = \begin{cases} 2\Big(1-0\Big)(1-y)\Big(1-0\Big)^{-3} = 2(1-y), & 0 < y < \frac{1}{3}, \\ 2\Big(1-\frac{3y-1}{2y}\Big)(1-y)\Big(1-\frac{3y-1}{2y}y\Big)^{-3} = \frac{8}{27}\frac{1}{y(1-y)}, & \frac{1}{3} < y < 1, \end{cases}$$

such that $f(x|y) \leq M(y)g(x)$ for all $0 < x < 1$ and $0 < y < 1$.

```
f = function(x, y){
  2*(1-x)*(1-y)*(1-x*y)^(-3)
}
g = function(x){
  dunif(x)
}
M = function(y){
```

```
  ifelse(0<y & y<1/3,2*(1-y),(8/27)*(1/(y*(1-y))))
}
```

2. Algorithm in details

Exact sampling and rejection sampling are combined in use. The steps are detailed as follows.

step1: Generate $Y \sim U(0,1)$.

step2: Generate $X \sim g$ and $U \sim U(0,1)$.

step3: Iterate step2 until $U \leq \frac{f(Y|X)}{M(Y)g(X)}$, and keep only the last X.

step4: Iterate step1 to step3 until 1000 pairs of $(X,Y)'s$ are generated throughout.

Note step1 is exact sampling and step2 to step3 is rejection sampling.

3. Draw a sample of size 1000.

```
x.vec = y.vec = c()
effs = c()
set.seed(111024513)
while(length(x.vec)<1000){
  y = runif(1)
  y.vec = c(y.vec, y)
  reject = TRUE
  n.gen = 0
  while(reject){
    x = runif(1)
    if(runif(1) <= f(x, y)/(M(y)*g(x))){
      x.vec = c(x.vec, x)
      reject = FALSE
    }
    n.gen = n.gen + 1
  }
  effs = c(effs, 1/n.gen)
}
```

4. compare the empirical distribution of your data to the target pdf.

Figure 8 shows that the empirical distribution looks very similar to the target distribution.

```
x = seq(0, 1, length.out = 100)
y = seq(0, 0.9999, length.out = 100)
z = outer(x, y, f)
image(x, y, log(z))
contour(x, y, z, levels= round(quantile(z, 0.05*(1:20)),2), add = TRUE)
points(x.vec, y.vec, pch = 20)
```

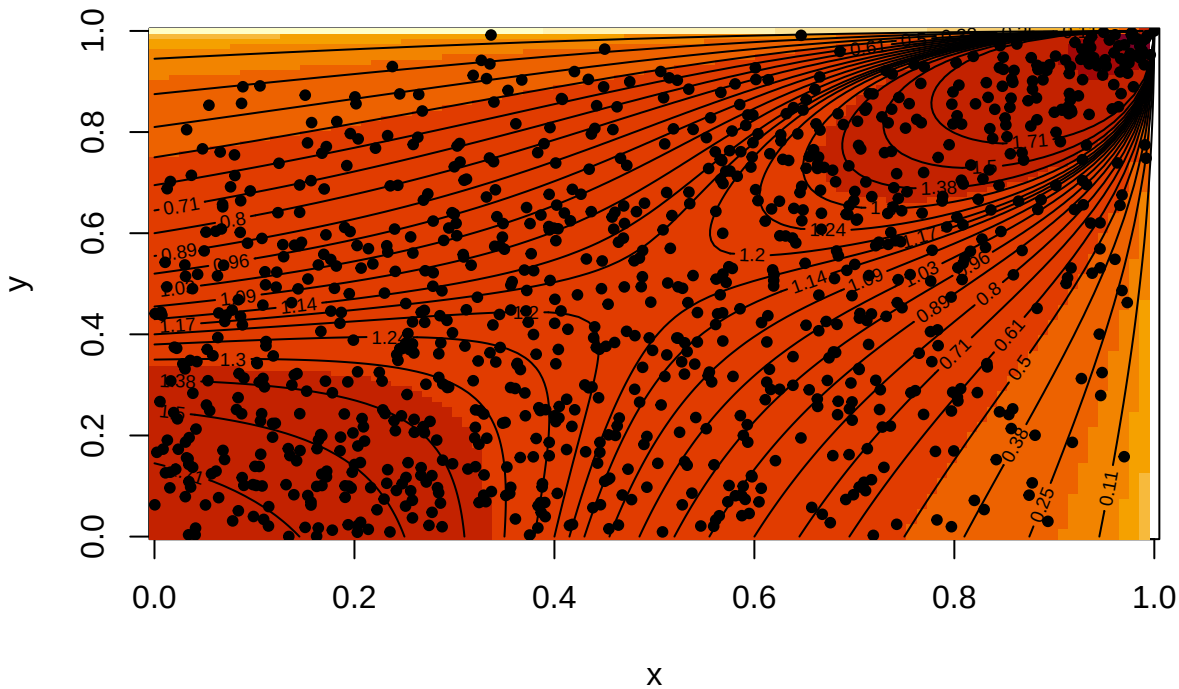
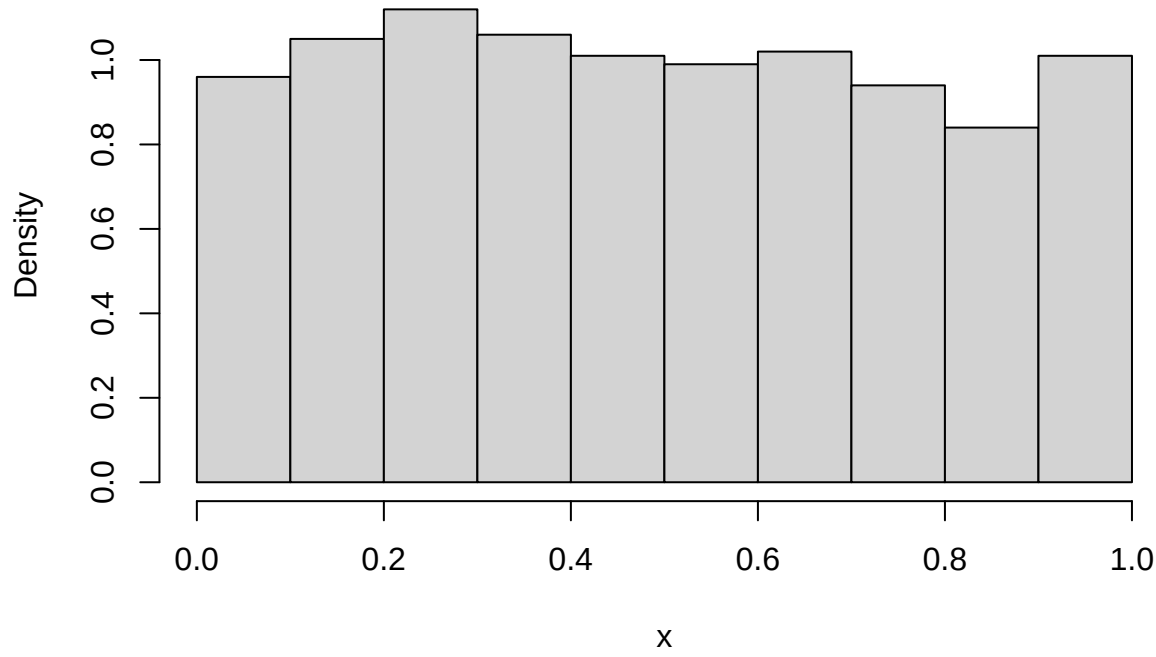


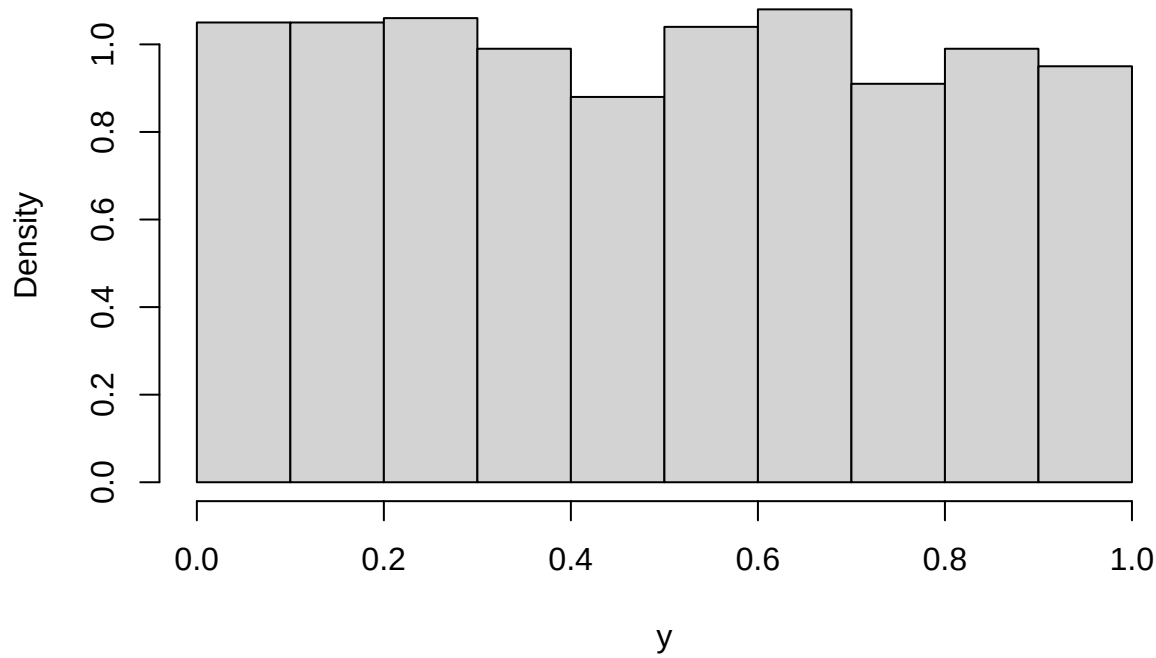
Figure 8: Emprical vs $f(x, y)$

Finally, the histograms below show the marginal distributions for X and Y, both of which look very close to uniform distribution.

```
hist(x.vec, xlab = "x", main = "", freq = FALSE)
```




```
hist(y.vec, xlab = "y", main = "", freq = FALSE)
```



5.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue.

```
mean(effs)
```

```
## [1] 0.7740101
```

The theoretical efficiency can be derived as follows.

$$
\begin{aligned}
\text{theoretical efficiency} &= E\Big\{\frac{1}{M(Y)}\Big\} \\
&= \int_0^{1/3}\int_0^1 \frac{f(x,y)}{M(y)}dxdy + \int_{1/3}^1\int_0^1 \frac{f(x,y)}{M(y)}dxdy \\
&= \int_0^{1/3}\int_0^1 \frac{(1-x)}{(1-xy)^3}dxdy + \frac{27}{4}\int_{1/3}^1 y(1-y)^2\int_0^1 \frac{(1-x)}{(1-xy)^3}dxdy \\
&= \frac{1}{2}\int_0^{1/3}\frac{1}{1-y}dy + \frac{27}{8}\int_{1/3}^1 y(1-y)dy \\
&= -\frac{1}{2}\cdot\ln\frac{2}{3} + \frac{27}{8}\cdot\frac{10}{81} \approx 61.94\%
\end{aligned}
$$

On the other hand, the empirical efficiency can be obtained by the formula below.

$$\text{empirical efficiency} = \frac{1}{1000}\sum_{i=1}^{1000}\frac{1}{N_{y_i}},$$

where N_{y_i} refers to the number of x generated from $f(x|y_i)$ for each given y_i generated from the uniform distribution. It turns out that the empirical efficiency is 77.401 %. We see that the empirical efficiency is larger than the theoretical one. This should be due to the fact that $\frac{1}{N_{y_i}}$ is an biased estimator for $\frac{1}{M(y_i)}$ which often leads to overestimation. Such result can be shown easily by noting that N_{y_i} follows a geometric distribution with success probability $p = \frac{1}{M(y_i)}$, and therefore

$$\begin{aligned} E\Big\{\frac{1}{N_{y_i}}\Big\} &= p\sum_{x=1}^{\infty}\frac{1}{x}(1-p)^{x-1} \\ &= p(1+\frac{1}{2}(1-p)^1+\frac{1}{3}(1-p)^2+\dots) > p. \end{aligned}$$

Note that the formula for empirical efficiency here is different in essence from that applied in previous problems. In fact, when applying the formula as used before, the empirical efficiency is only around 10.89 %. An alternative approach is provided in appendix, in which two-dimensional rejection sampling is applied. It is also shown that this alternative approach provides higher empirical efficiency in terms of the formula as used before.

(TA may skip the appendix since R function `optim` which may not be allowed is used. The appendix is placed here only for my self practice.)

6 Acknowledgement

Thanks to Mr.Kang (康暐郡) and Mr.Kuo (郭鎧任) for their valuable suggestions in problem 5, where they provided me the idea of factoring $f(x,y)$ into $f(y|x)f(y)$ and several derivation results.

7 Appendix

An alternative approach for problem 5 is shown as follows.

7.1 Draw a sample of size 1000 for the distribution with a specific parameter value and compare the empirical distribution of your data to the target pdf.

1. Find a suitable proposal distribution

In hope of grabing the overall shape of the target distribution, two 3-dimensional plots with different perspectives are shown in Figure 9. We can see that $f(x)$ is not a bounded function since it seems to take value arbitrarily large at points near $(0,0)$. In fact, we can easily obtain such finding theoretically as follows.

$$\lim_{x\to 0+,\, y\to 0+} f(x,y) = \lim_{x\to 0+,\, y\to 0+} 2(1-x)(1-y)(1-xy)^{-3} = \infty$$

This implies that rejection sampling as we used in the previous problems could be quite challenging to apply here since there may not be a well-known distribution with density $g(x,y)$ such that there exists M letting $f(x,y) \le Mg(x,y)$ for all $x > 0$.

However, instead of generating random numbers directly from $f(x)$, it may be a good idea to first do some transformation $T(\cdot,\cdot)$ on x and y which leads to a bounded density function on the new support. Then apply rejection sampling on such "new" target distribution to generate a set of random numbers. After that, we can transform these random numbers back via $T^{-1}(\cdot,\cdot)$ and thereby obtain the desired random numbers from the "original" target distribution.

```
f.org = function(x, y){
  2*(1-x)*(1-y)*(1-x*y)^(-3)
}
```

```
x = y = seq(0, 1, length.out = 100)
z = outer(x, y, f.org)
par(mfrow = c(1, 2))
persp(x, y, z, theta = 120, col = "skyblue")
persp(x, y, z, theta = 240, col = "skyblue")
```

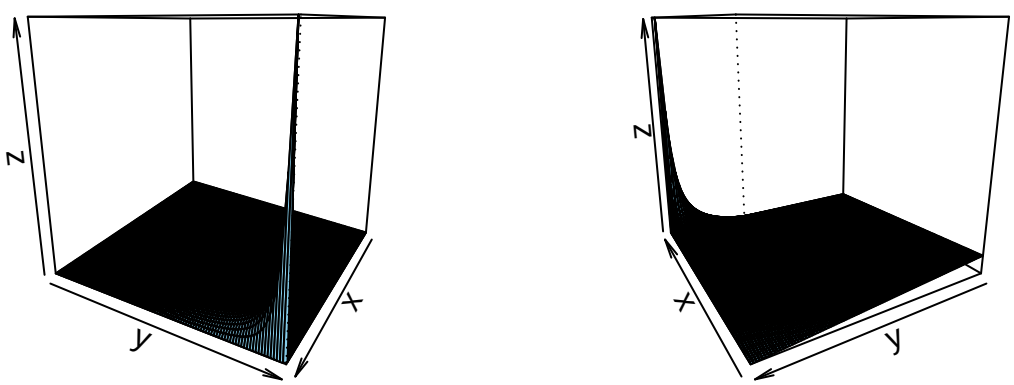


Figure 9: Visualization of $f(x, y)$

It turns out that by letting $u = \sqrt{-\log(x)}$ and $v = \sqrt{-\log(y)}$ with the associated Jacobian,

$$J = \begin{pmatrix} -2ue^{-u^2} & 0 \\ 0 & -2ve^{-v^2} \end{pmatrix} = 4uve^{-u^2-v^2},$$

the density function for the new target distribution $f^*(x, y)$ can be derived as

$$
\begin{aligned}
f^*(u,v) &= 2(1-e^{-u^2})(1-e^{-v^2})(1-e^{-u^2-v^2})^{-3}|J| \\
&= 8uv\frac{e^{-u^2-v^2}}{1-e^{-u^2-v^2}}\frac{1-e^{-u^2}}{1-e^{-u^2-v^2}}\frac{1-e^{-v^2}}{1-e^{-u^2-v^2}}, \quad 0<u<\infty, \quad 0<v<\infty
\end{aligned}
$$

Figure 10 gives an intuition regarding the shape of $f^*(x,y)$. It seems more likely to find a proposal distribution suitable for rejection sampling for such target distribution.

```
f = function(x, y){
  8*(1-exp(-x^2))*(1-exp(-y^2))*((1-exp(-x^2-y^2))^(-3))*x*y*exp(-x^2-y^2)
}
g = function(x,y){
  dexp(x)*dexp(y)
}
```

```
x = y = seq(0, 4, length.out = 100)
z = outer(x, y, f)
par(mfrow = c(1, 2))
persp(x, y, z, theta = 120, col = "skyblue")
persp(x, y, z, theta = 300, col = "skyblue")
```

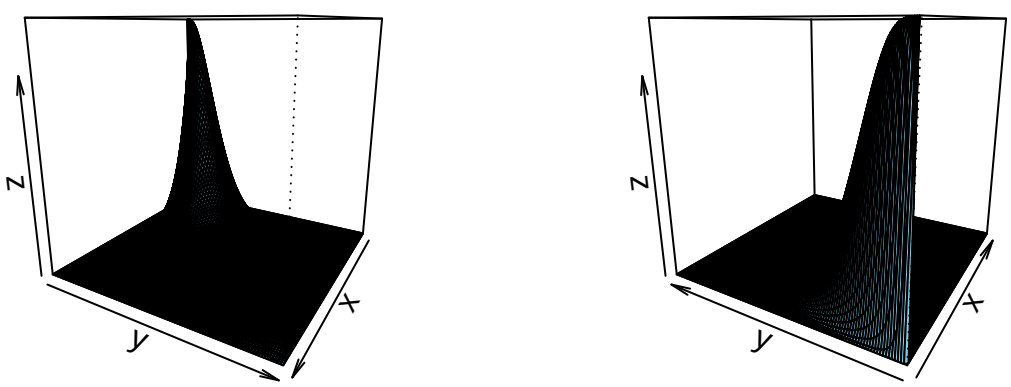


Figure 10: Visualization of $f^*(x,y)$

Consider $g(u,v) = e^{-u^2-v^2}$ which is the density of the joint distribution for two independent exponential

variables. It suffices to examine whether $\frac{f^*(x,y)}{g(x,y)}$ is bounded on $R^+ \times R^+$. Letting $u \to 0+$ and $v \to 0+$, we have

$$\begin{aligned}
\lim_{u\to 0+,\, v\to 0+} \frac{f^*(u,v)}{g(u,v)} &= \lim_{u\to 0+,\, v\to 0+} \frac{8uv}{e^{-u-v}} \frac{e^{-u^2-v^2}}{1-e^{-u^2-v^2}} \frac{1-e^{-u^2}}{1-e^{-u^2-v^2}} \frac{1-e^{-v^2}}{1-e^{-u^2-v^2}} \\
&= \lim_{u\to 0+,\, v\to 0+} \frac{8}{e^{-u-v}} \frac{1}{\frac{1+(u^2+v^2)+(u^2+v^2)^2+\dots}{uv} - 1} \frac{u^2-u^4+\dots}{(u^2+v^2)-(u^2+v^2)^2+\dots} \frac{v^2-v^4+\dots}{(u^2+v^2)-(u^2+v^2)^2+\dots} \\
&= 8 \cdot 0 \cdot 1 \cdot 1 = 0.
\end{aligned}$$

The second equality follows by Taylor's theorem. On the other hand letting $u \to \infty$ and $v \to \infty$, we have

$$\begin{aligned}
lim_{u\to\infty,\, v\to\infty} \frac{f^*(u,v)}{g(u,v)} &= \lim_{u\to\infty,\, v\to\infty} \frac{8uv}{e^{-u-v}} \frac{e^{-u^2-v^2}}{1-e^{-u^2-v^2}} \frac{1-e^{-u^2}}{1-e^{-u^2-v^2}} \frac{1-e^{-v^2}}{1-e^{-u^2-v^2}} \\
&= \lim_{u\to\infty,\, v\to\infty} \frac{8uve^{u+v}}{e^{u^2+v^2}-1} \frac{1-e^{-u^2}}{1-e^{-u^2-v^2}} \frac{1-e^{-v^2}}{1-e^{-u^2-v^2}} \\
&= \lim_{u\to\infty,\, v\to\infty} \frac{8uv(1+(u+v)+(u+v)^2+(u+v)^3\dots)}{(u^2+v^2)+(u^2+v^2)^2+(u^2+v^2)^3\dots} \frac{1-e^{-u^2}}{1-e^{-u^2-v^2}} \frac{1-e^{-v^2}}{1-e^{-u^2-v^2}} \\
&= 0 \cdot 1 \cdot 1 = 0.
\end{aligned}$$

The Third equality follows by Taylor's theorem. By noting that $\frac{f^*(u,v)}{g(u,v)}$ is a continuous function on $R^+ \times R^+$, it follows that it is bounded on $R^+ \times R^+$. Hence, by Max-Min Theorem and noting that $\frac{f^*(u,v)}{g(u,v)}$ is non-negative, there must exist a maximum on $R^+ \times R^+$, and thus we can find $M = \max_{u>0,v>0} \frac{f^*(u,v)}{g(u,v)}$. Figure 11 shows the shape of $\frac{f^*(u,v)}{g(u,v)}$.

```
x = y = seq(0, 4, length.out = 100)
z = outer(x, y, function(x,y) f(x,y)/g(x,y))
par(mfrow = c(1, 2))
persp(x, y, z, theta = 120, col = "skyblue")
persp(x, y, z, theta = 300, col = "skyblue")
```

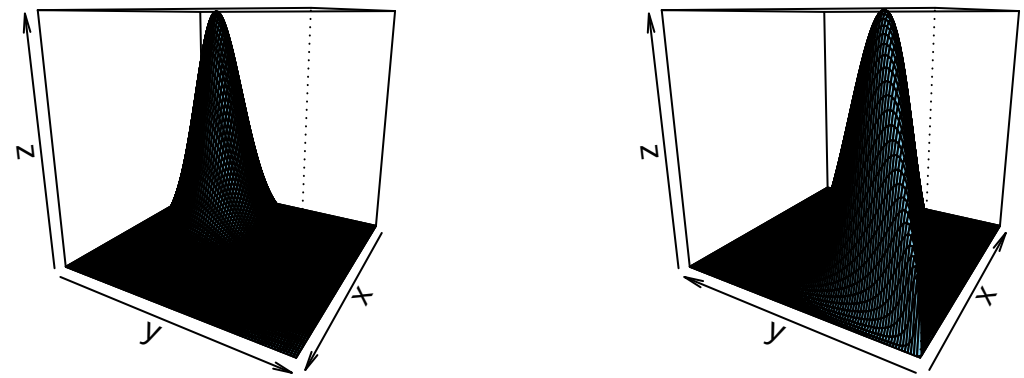


Figure 11: Visualization of $f^*(x, y)$

```
obj = function(pars){
  -f(pars[1], pars[2])/g(pars[1], pars[2])
}
opt = optim(c(0.1, 0.1), obj, method = "L-BFGS-B",
            lower = c(10^(-8), 10^(-8)), upper  = c(3,4))
M = -opt$value
```

By using R function `optim`, we can numerically obtain $M = 5.0593352$.

2. Algorithm in details

Rejection sampling is applied. The steps are detailed as follows.

step1: Generate $(X, Y) \sim g$ and $U \sim U(0, 1)$.

step2: Iterate step1 until $U \leq \frac{f(X,Y)}{Mg(X,Y)}$, and keep only the last pair of (X, Y).

step3: Iterate step1 and step2 until 1000 pairs of $(X, Y)'s$ are generated throughout.

3. Draw a sample of size 1000**.**

```
x.vec = y.vec =  c()
n.gen = 0
set.seed(111024513)
while(length(x.vec)<1000){
  x = rexp(1); y = rexp(1)
```

```
  if(runif(1) <= f(x, y)/(M*g(x,y))){
    x.vec = c(x.vec, x)
    y.vec = c(y.vec, y)
  }
  n.gen = n.gen + 1
}
```

4. compare the empirical distribution of your data to the target pdf.

Figure 12 shows that the empirical distribution looks very similar to $f^*(u, v)$.

```
x = y = seq(0, 3, length.out = 100)
z = outer(x, y, f)
image(x,y,z, xlab = expression(u), ylab = expression(v))
contour(x, y, z, add = TRUE)
points(x.vec, y.vec)
```

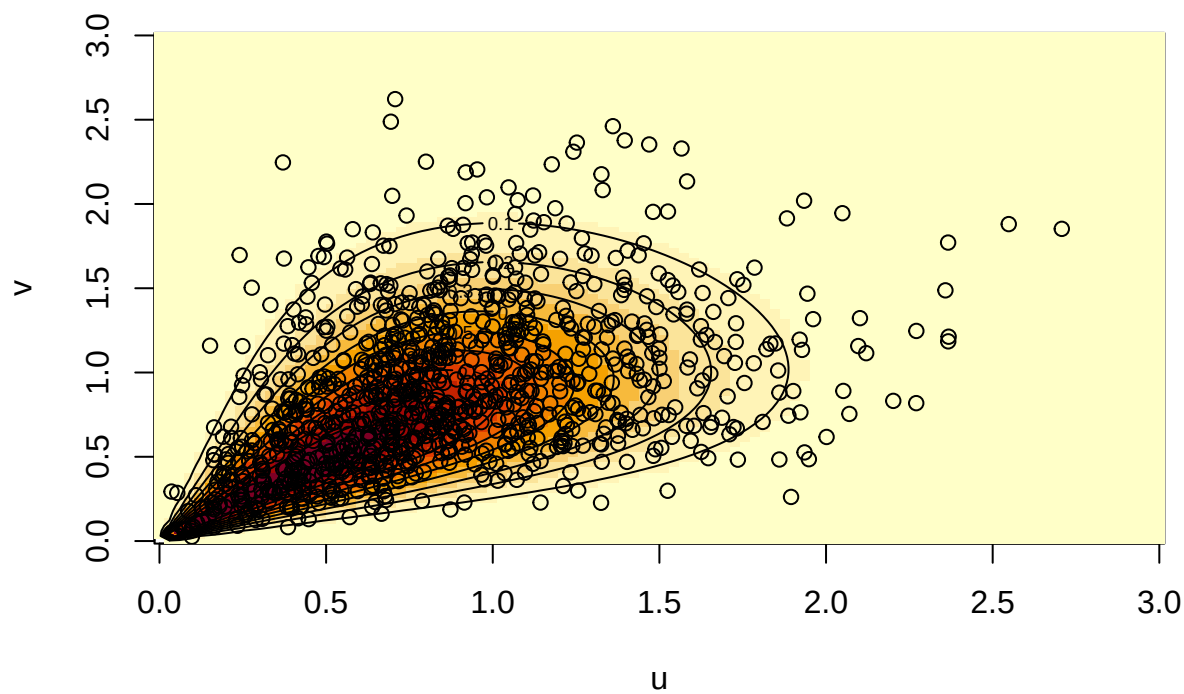


Figure 12: Emprical vs $f^*(u, v)$

By taking the inverse transformation on these random numbers, we can obtain the ones that come from the "orignal" target distribution. Figure 13 shows that the empirical distribution of the transformed numbers looks very similar to $f(x, y)$.

```
x = seq(0, 1, length.out = 100)
y = seq(0, 0.9999, length.out = 100)
z = outer(x, y, f.org)
image(x, y, log(z))
contour(x, y, z, levels= round(quantile(z, 0.05*(1:20)),2), add = TRUE)
points(exp(-x.vec^2), exp(-y.vec^2), pch = 20)
```

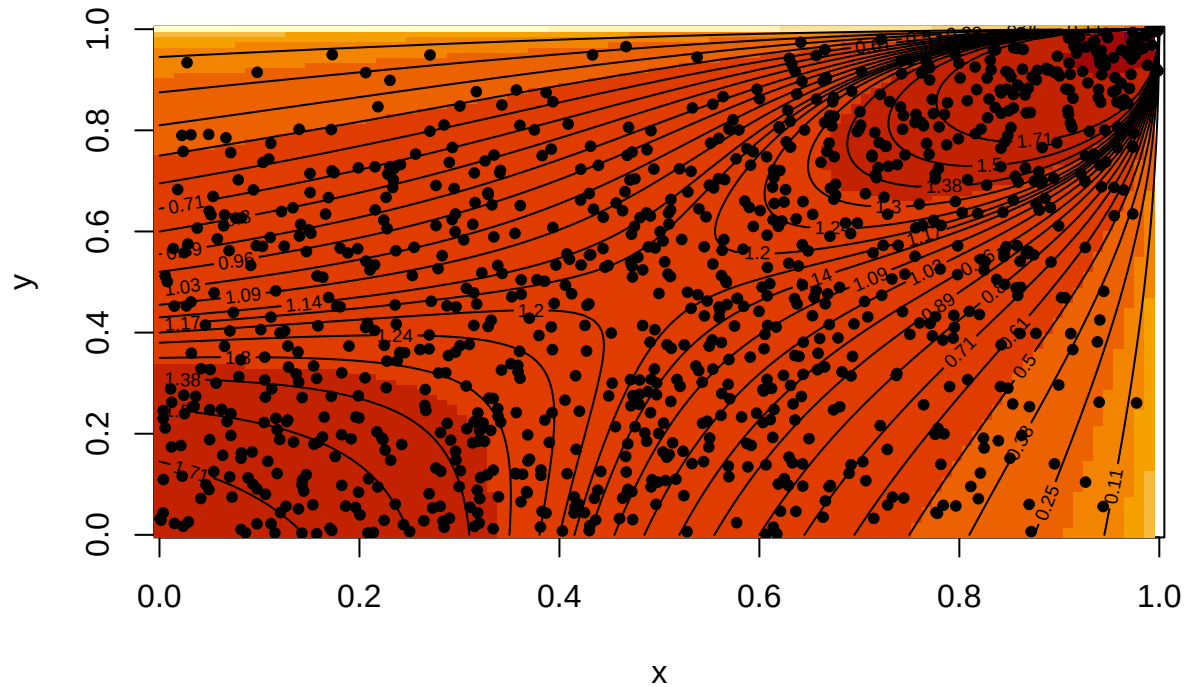


Figure 13: Emprical vs $f(x, y)$

7.2 Evaluate the efficiency (theoretically and empirically) of your sampling algorithm. Does the efficiency change with the parameter values? Make some comments on this issue.

```
1000/n.gen
```

```
## [1] 0.1916076
```

The theoretical efficiency is $\frac{1}{M} = 19.7654$ % and the empirical efficiency can be obtained by dividing 1000 to the number of the random numbers generated from $g(\cdot)$. It turns out that the empirical efficiency is 19.1608 %. We see that the empirical efficiency is very close to the theoretical one as expected.